Jongyul Kim

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RESEARCH INTERESTS

My research investigates how system software should be re-architected at scale when assumptions about workloads and hardware no longer hold. My work spans storage and memory systems, distributed and disaggregated architectures, smart devices such as SmartNICs and computational storage devices, and Compute Express Link (CXL), with growing interest in system support for AI/ML workloads.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea

Ph.D. (Integrated M.S./Ph.D. Program), School of Computing — Feb 2022

Dissertation: "Distributed Persistent Memory File System for Programmable NIC"

Advisors: Seungryoul Maeng and Youngjin Kwon

B.S. in Computer Science and Management Science (Double Major) — Feb 2012

AWARDS

- Best Paper Award (SOSP 2021), ACM SIGOPS 28th Symposium on Operating Systems Principles, 2021.
- Best Dissertation Award, School of Computing, KAIST, 2022.
- KAIST Breakthroughs (Spring 2022), Selected Research Highlight, Biannual Research Webzine, 2022.
- Best Teaching Assistant Award (Fall 2014), School of Computing, KAIST, 2015.
- Best Teaching Assistant Award (Fall 2013), School of Computing, KAIST, 2014.

PUBLICATIONS

[1] Minkyu Jung, Chanshin Kwak, Junho Ahn, Sunho Park, Changjun Lee, **Jongyul Kim**, Jeehoon Kang, and Youngjin Kwon. "CofferOS: Hardening OS-level Virtualization with Rust". *Proceedings of the Twenty-First European Conference on Computer Systems.* (EuroSys 2026).

[2] Siyuan Chai, Jiyuan Zhang, **Jongyul Kim**, Alan Wang, Fan Chung, Jovan Stojkovic, Weiwei Jia, Dimitrios Skarlatos, Josep Torrellas, and Tianyin Xu. "EMT: An OS Framework for New Memory Translation Architectures". *Proceedings of the 19th USENIX Symposium on Operating Systems Design and Implementation.* (OSDI 2025).

[3] Jiyuan Zhang, **Jongyul Kim**, Chloe Alverti, Peizhe Liu, Weiwei Jia, and Tianyin Xu. "Rethinking Tiered Storage: Talk to File Systems, Not Device Drivers". *Proceedings of the 20th Workshop on Hot Topics in Operating Systems.* (HotOS 2025).

[4] Yan Sun, **Jongyul Kim**, Douglas Yu, Jiyuan Zhang, Siyuan Chai, Jaemin M. Kim, Hwayong Nam, Jaehyun Park, Eojin Na, Yifan Yuan, Ren Wang, Jung Ho Ahn, Tianyin Xu, and Nam Sung Kim. "M5: Mastering Page Migration and Memory Management for CXL-based Tiered Memory Systems". *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems.* (ASPLOS 2025).

[5] Jiyuan Zhang, Weiwei Jia, Siyuan Chai, Peizhe Liu, **Jongyul Kim**, and Tianyin Xu. "Direct Memory Translation for Virtualized Clouds". *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems.* (ASPLOS 2024).

[6] **Jongyul Kim**, Insu Jang, Waleed Reda, Jaeseong Im, Marco Canini, Dejan Kostić, Youngjin Kwon, Simon Peter, and Emmett Witchel. "LineFS: Efficient SmartNIC Offload of a Distributed File System with Pipeline Parallelism". *13th Annual Non-Volatile Memories Workshop 2022.* (NVMW 2022).

[7] Jaeseong Im, **Jongyul Kim**, Youngjin Kwon, and Seungryoul Maeng. "On-demand Virtualization for Post-copy OS Migration in Bare-metal Cloud". *IEEE Transactions on Cloud Computing.* 2022. **Impact factor: 5.938** by WOS.

[8] **Jongyul Kim**, Insu Jang, Waleed Reda, Jaeseong Im, Marco Canini, Dejan Kostić, Youngjin Kwon, Simon Peter, and Emmett Witchel. "LineFS: Efficient SmartNIC Offload of a Distributed File System with Pipeline Parallelism". *Proceedings of the ACM SIGOPS 28th Symposium on Operating Systems Principles.* **Best Paper Award.** (SOSP 2021).

[9] Thomas E. Anderson, Marco Canini, **Jongyul Kim**, Dejan Kostić, Youngjin Kwon, Simon Peter, Waleed Reda, Henry N. Schuh, and Emmett Witchel. "Assise: Performance and Availability via Client-local NVM in a Distributed File System". *12th Annual Non-Volatile Memories Workshop 2021.* **Co-student author, Memorable Paper Award Finalists**. (NVMW 2021).

[10] Thomas E. Anderson, Marco Canini, **Jongyul Kim**, Dejan Kostić, Youngjin Kwon, Simon Peter, Waleed Reda, Henry N. Schuh, and Emmett Witchel. "Assise: Performance and Availability via Client-local NVM in a Distributed File System". *Proceedings of the 14th USENIX Symposium on Operating Systems Design and Implementation.* **Co-student author**. (OSDI 2020).

[11] Jaeseong Im, **Jongyul Kim**, Jonguk Kim, Seongwook Jin, and Seungryoul Maeng. "On-demand Virtualization for Live Migration in Bare Metal Cloud". *Proceedings of the 2017 Symposium on Cloud Computing.* (SoCC 2017).

[12] Jaeseong Im, **Jongyul Kim**, and Seungryoul Maeng. "Whole System Checkpoint-recovery Mechanism in Bare-metal In-memory System". *Korea Computer Congress 2017.* (KCC 2017).

RESEARCH EXPERIENCE

University of Illinois Urbana-Champaign (UIUC) **Urbana, IL, USA**

- **Postdoctoral Research Associate**

  A cross-layer file system architecture with orchestrated modular components [13] (Sep 2023 – present)
  Re-architected a file system to overcome the limitations of monolithic designs by decomposing and orchestrating file system components across user space, the Linux kernel, and programmable storage devices.
  Technical skills: File system architecture, Linux kernel, SPDK, DPU (Data Processing Unit), RDMA, NVMe-oF

  An extensible abstraction layer for composing modular tiered file systems [3] (Oct 2023 – Aug 2025)
  Designed an extensible abstraction for composing heterogeneous, device-optimized file systems into a unified file system view.
  Heterogeneous storage devices, File system abstraction

  An OS-level framework for diverse memory translation architectures [2] (Oct 2023 – Jun 2025)
  Enabled systematic design and evaluation of alternative memory translation mechanisms by decoupling OS memory management from hardware-defined translation structures, via an OS abstraction and toolchains.
  Linux kernel, Memory translation

  CXL-based tiered memory management with on-device access tracking [4] (Oct 2023 – Mar 2025)
  Developed an OS-driven tiered memory management mechanism that leverages hotness profiling logic implemented on a CXL memory device.
  CXL, Memory tiering, Linux kernel, Hot page detection, Page migration

  Direct memory translation mechanism [5] (Oct 2023 – Apr 2024)
  Proposed a direct memory translation mechanism that bypasses conventional page-based address translation for both native and virtualized environments.
  Linux kernel, Memory management, Page tables, Virtualization

Korea Advanced Institute of Science and Technology (KAIST) **Daejeon, South Korea**

- **Postdoctoral Researcher**

  A user-level file system that leverages computational storage devices (Oct 2022 – Sep 2023)
  Designed a user-level file system that reduces host CPU utilization by offloading I/O functionality to computational storage devices.
  User-level file system architecture, SPDK, Computational storage, DPU, RDMA, NVMe-oF

  In-SmartNIC metadata cache for Lustre client (May 2022 – Dec 2022)
  Improved Lustre metadata scalability and lookup latency by serving metadata requests from SmartNIC DRAM, bypassing the metadata server.
  Lustre, SmartNIC (DPU), RDMA, Metadata management

  Rack-scale RDMA-based memory disaggregation (Dec 2021 – May 2022)
  Implemented a disaggregated memory system by extending the Linux kernel swap mechanism over RDMA.
  Memory disaggregation, Linux kernel, Memory management, RDMA

- **Graduate Researcher**

  Distributed file system for persistent memory with SmartNIC offload [8, 6] (Jun 2019 – Jan 2022)
  Developed a rack-scale distributed file system that mitigates interference in node-local persistent memory systems by offloading data-path operations to SmartNICs.

Distributed file system, SmartNIC, Persistent memory, RDMA, HW Acceleration, Parallel processing

Distributed file system for persistent memory over RDMA [10, 9] (Oct 2018 – Oct 2020)
Proposed a distributed file system design tailored to node-local persistent memory architectures in rack-scale environments.
Distributed file system architecture, Persistent memory, RDMA, NFS, Ceph

On-demand virtualization for bare-metal cloud [11, 7] (Jul 2015 – Sep 2018)
Proposed an on-demand virtualization mechanism that dynamically virtualizes bare-metal instances at run-time, enabling live migration and checkpointing.
Virtualization, Cloud, KVM, Live migration, Checkpointing

Memory-centric architecture with processing-in-memory (Mar 2016 – Oct 2018)
Explored memory-centric architectures with processing-in-memory (PIM) using Hybrid Memory Cube (HMC) by simulating multi-HMC systems with memory management offloaded to PIM.
PIM, Memory-centric architecture, HMC, gem5, McSimA+

WORK EXPERIENCE

Postdoctoral Research Associate, University of Illinois Urbana-Champaign — Urbana, IL, USA
Siebel School of Computing and Data Science, Host: Prof. Tianyin Xu. — Sep 2023 – present

Postdoctoral Researcher, KAIST — Daejeon, South Korea
School of Computing, Host: Prof. Youngjin Kwon. — Mar 2022 – Sep 2023

Startup Co-Founder / Software Developer, Durooh, Inc. — Seoul, South Korea
Led feature planning and developed an Android application for an early-stage product. — Jun 2011 – Feb 2013

Undergraduate Intern, TestMidas Co., Ltd. — Daejeon, South Korea
Ported Windows system calls to Linux by analyzing the *Wine* codebase. — Jun 2009 – Aug 2009

MENTORING

Mentoring at UIUC

- Peizhe Liu — Dec 2024 – present
  Developed an efficient file system journal checkpointing scheme on a DPU.

Mentoring at KAIST

- Donggeun Kim — Jan 2022 – Aug 2022
  Continued the development of a hash-based file mapping in a persistent-memory file system.
- Guseul Heo — Aug 2021 – Dec 2021
  Replaced the extent tree with a hash-based file mapping in a persistent-memory file system.
- Jaehwan Lee — Aug 2021 – Dec 2021
  Added multi-threading support to a persistent-memory file system.

TEACHING EXPERIENCE

Teaching at UIUC

- Operating System Design (CS423, Honorary Instructor) — Fall 2024
- UIUC Systems Research Seminar (CS591, Co-host) — Spring 2024 – Fall 2025

Teaching Assistant at KAIST

- Digital System and Lab (CS211) — Spring 2014 (Head), Spring 2015 (Head)
  Lab sessions: VHDL (Hardware description language) programming.
- Embedded Computer Systems (CS310) — Fall 2013 (Head), Fall 2014, Fall 2015
  Lab sessions: FPGA and Arduino microcontroller programming.
- Embedded Computing (SEP561) — Spring 2014 (Head), Spring 2015, Spring 2019
  Lab sessions: FPGA and Arduino microcontroller programming.

SERVICE

Program Committee Member

- USENIX ATC: 2025.

External Program Committee

- USENIX ATC: 2024.

Shadow PC

- EuroSys: 2023.

Journal Reviewer

- ACM Transactions on Storage: 2022, 2025.

INVITED TALKS

- "LineFS: Efficient SmartNIC Offload of a Distributed File System with Pipeline Parallelism", UIUC Systems Research Seminar, Urbana, IL, US, January 2024.
- "Persistent-memory-based Distributed File System and SmartNIC Offloading", *Electronics and Telecommunications Research Institute (ETRI)* Seminar, Daejeon, South Korea, Mar 2023.
- "Persistent-memory-based Distributed File System and SmartNIC Offloading", *Electronic & Information Research Information Center (EIRIC)* Seminar, Virtual, June 2022.
- "LineFS: Efficient SmartNIC Offload of a Distributed File System with Pipeline Parallelism", Top conference session in *Korea Software Congress 2021 (KSC 2021)*, Pyeongchang, Gangwon, South Korea, December 2021.

PATENTS & APPLICATIONS

1. Patent Application (KO/US/CN/EP), 10-2022-0173904, "COMPUTABLE NETWORK INTERFACE CARD AND ELECTRONIC APPARATUS INCLUDING THE SAME", Dec 2022.
2. Patent Application (KO), 10-2022-0165086, "METHOD FOR MANAGING MEMORY AND COMPUTER DEVICE FOR THE SAME", Nov 2022.

SKILLS

DEVELOPMENT

- C, C++, Java, Python, Shell, JavaScript, Linux kernel development
- File systems, Distributed file systems, Memory management (including disaggregation, tiering), Virtualization (KVM), User-space I/O (SPDK)
- RDMA, Persistent memory, SSDs, CXL, SoC-based smart devices (SmartNICs/DPUs & SmartSSDs), Hardware acceleration (DMA engines)
- gem5 simulator, Android

LANGUAGES

English, Korean